



VEHICLE ACCEPTABILITY DETERMINATION

GREEN, MIKE



Executive Summary

This report is made with the goal of determining a car's acceptability based on its specific features. The analysis evaluates both the price and comfort of each vehicle, specifically focusing on the buying price, maintenance cost, number of doors, capacity, the size of the luggage boot, and the safety of each car.

Based on these features, each vehicle is assigned a hierarchical classification label from worst to best: unacceptable, acceptable, good, and very good. This model aims to assist potential buyers in narrowing down their options when making a significant purchase decision.

The results demonstrate strong performance across the two classification models used, Random Forest and Decision Tree, with both achieving over 98% accuracy and 99% precision. The weighted F1 score of 0.99 further indicates the model performs consistently well across all four classification labels, giving customers further peace of mind using our model.

Data Preprocessing

The *Car Evaluation Dataset* by Esther Kanyi was loaded into Python for analysis. Because the dataset did not include predefined column headers, feature names were manually assigned: buying, maintenance, doors, persons, lug_boot, safety, and class.

The dataset was imported using the Pandas `read_csv` functions with the `names` parameter to apply the manually defined headers. After verifying that the data was correctly loaded, built-in functions were used to confirm the absence of missing values and to validate the data types of each feature.

Categorical features were then encoded into numerical form using `LabelEncoder` function from `sklearn`. The features *doors* and *persons* were excluded from encoding, as they were already represented as integers. Finally, the target variable was defined as the class label, which represents four possible hierarchical outcomes: unacceptable, acceptable, good, and very good.

```
Missing values for each feature:
buying      0
maintenance 0
doors       0
persons     0
lug_boot    0
safety      0
class       0
dtype: int64
Data types:
buying      str
maintenance str
doors       int64
persons     int64
lug_boot    str
safety      str
class       str
dtype: object
```

Exploratory Analysis of the data

The dataset contains 1,727 objects and 7 features, including the target class label. Among these features, *doors* and *persons* are represented as integer data types; while *buying*, *maintenance*, *lug_boot*, *safety*, and *class* are all categorical string variables.

	buying	maintenance	doors	persons	lug_boot	safety	class
0	vhigh	vhigh	2	2	small	med	unacc
1	vhigh	vhigh	2	2	small	high	unacc
2	vhigh	vhigh	2	2	med	low	unacc
3	vhigh	vhigh	2	2	med	med	unacc
4	vhigh	vhigh	2	2	med	high	unacc

Row Labels	Count of Column7
acc	384
good	69
unacc	1209
vgood	65
Grand Total	1727

The distribution of features values is generally balanced across categories. However, the target variable is notably imbalanced, with approximately 70% of the observations, 1,209 of the 1,727 entries, classified as unacceptable. This suggests that the model applies relatively strict criteria when determining vehicle acceptability for potential buyers.

Methods

To develop the model, the dataset was split into training and testing sets using an 80/20 ratio. Due to the class imbalance, the stratify parameter was applied to ensure that each split maintained proportional representation of all class labels. A random state parameter was also specified to ensure reproducibility of results, forcing the same split to be run every time.

```
X_train, X_test, y_train, y_test = train_test_split(X,y, test_size=0.2, random_state=42, stratify=y)
```

The training and testing data were then transformed so that features are centered around 0 and scaled consistently. This was accomplished using the StandardScaler function from sklearn, ensuring that all features contribute equally during model training.

With the data prepared, a Random Forest Classifier was implemented using 100 trees and a fixed random state. This model constructs multiple decision trees, each trained on different portions of the data, and combines their results to make a final prediction. The final classification is determined by a majority voting among the decision trees, with the most frequent prediction selected as the output.

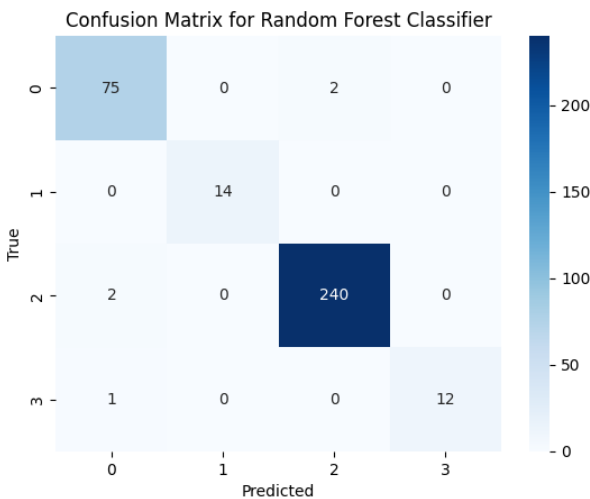
For comparison, a Decision Tree Classifier was also implemented. This model operates similarly to a flow chart, following paths based on each feature's values to arrive at a final classification.

Analysis of Results

Both models produced strong results, with accuracy scores of 98.55% and 98.84% for the Random Forest and Decision Tree classifiers respectively. This indicates that over 98% of predictions on the test set were correctly classified.

However, the F1 score is a more appropriate metric for this dataset due to the class imbalance, with *unacceptable* dominating the distribution. Since the F1 score incorporates both precision and recall, it provides a more balanced evaluation of performance across all class labels. Both models achieved a weighted F1 score of 0.99, indicating consistent and reliable performance across all four classifications.

Random Forest Classifier Results:

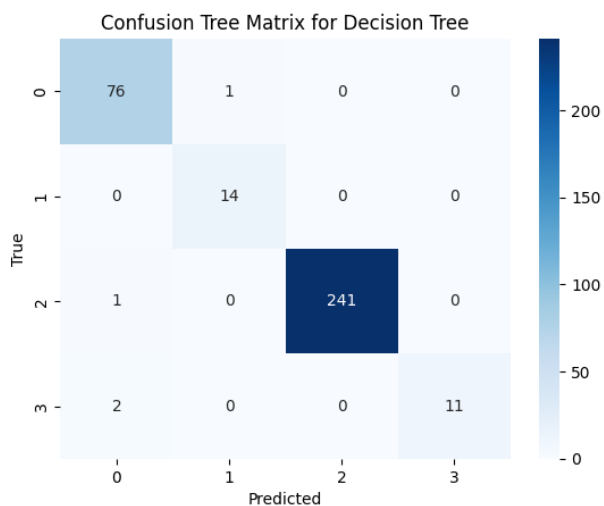


```
Random ForestAccuracy: 0.9855491329479769
Random Forest Classification Report:
              precision    recall  f1-score   support

     acc       0.96       0.97       0.97        77
    good       1.00       1.00       1.00        14
   unacc       0.99       0.99       0.99       242
    vgood       1.00       0.92       0.96        13

 accuracy                   0.99       346
 macro avg       0.99       0.97       0.98       346
weighted avg       0.99       0.99       0.99       346
```

Decision Tree Classifier Results:



```
Decision Tree Accuracy: 0.9884393063583815
Decision Tree Classification Report:
              precision    recall  f1-score   support

     acc       0.96       0.99       0.97        77
    good       0.93       1.00       0.97        14
   unacc       1.00       1.00       1.00       242
    vgood       1.00       0.85       0.92        13

 accuracy                   0.99       346
 macro avg       0.97       0.96       0.96       346
weighted avg       0.99       0.99       0.99       346
```

Validation and Testing

To ensure the reliability of the results, several validation measures were applied throughout the modeling process. The dataset was split using an 80/20 ratio, with 20% used for testing on observations not seen during training. The stratify parameter was used during the split to maintain proportional class representation in both the training and testing sets, which was important given the imbalance. A fixed random state was also implemented to ensure reproducibility and consistency of results.

Multiple evaluation metrics were recorded to provide a comprehensive picture of model performance, including accuracy, precision, recall, and F1 score. Additionally, a second model was created for comparison to confirm the strong results were not specific to a single classifier. Both the Random Forest and Decision Tree models performed above 98%.

Discussion and Conclusions

The models created performed exceptionally well and could be very useful for customers when purchasing a vehicle. With a weighted F1 score of 0.99 across both models, customers can feel confident in the classification of quality for their potential purchases based on the features used in the models.

While the models performed well, there are still some limitations to consider. The models focus on the basic aspects of car buying and do not account for factors such as mileage, brand, color, or other specific features that may be important to a customer. If explored further, additional features such as mileage and brand could be incorporated to provide a more comprehensive evaluation, potentially changing the acceptability rating.

Additionally, while the Decision Tree slightly outperformed the Random Forest in accuracy, both models still produced consistent results above 98%. This suggests the dataset is well-structured and that the selected features are strong predictors of vehicle acceptability. Despite the simplicity of the models, they are still effective and can help customers feel more confident in getting the most out of their purchase.

```
Random Forest Accuracy: 0.9855491329479769  
Decision Tree Accuracy: 0.9884393063583815
```

References

E. Kanyi, "Car Evaluation Dataset," *Kaggle*, [Online]. Available: <https://www.kaggle.com/datasets/kanyianalyst/car-evaluation-dataset>. [Accessed: Apr. 2025].